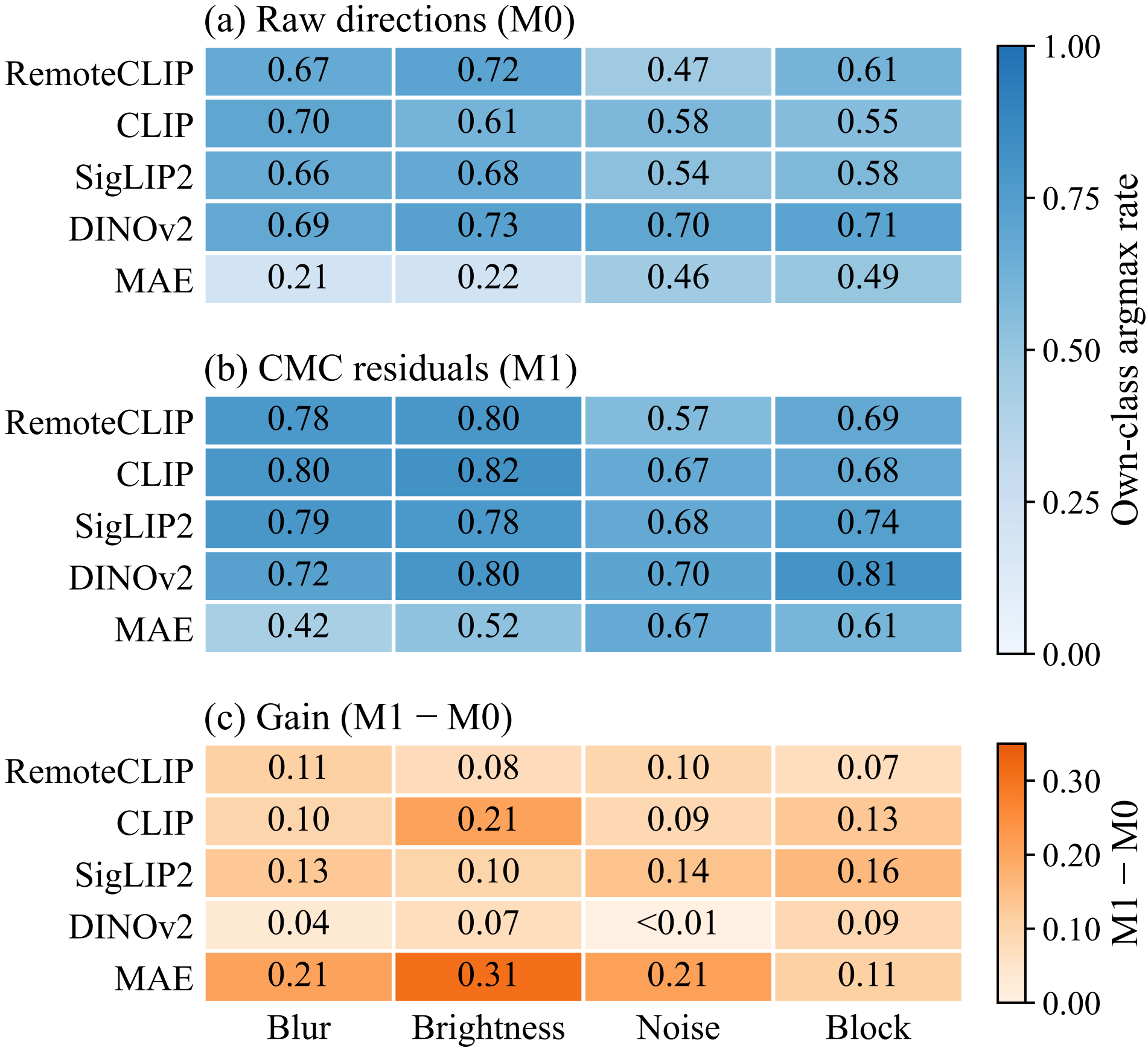
(a) Raw directions (M0)
RemoteCLIP 0.67 0.72 0.47 0.61
CLIP 0.70 0.61 0.58 0.55
SigLIP2 0.66 0.68 0.54 0.58
DINOv2 0.69 0.73 0.70 0.71
MAE 0.21 0.22 0.46 0.49
(b) CMC residuals (M1)
RemoteCLIP 0.78 0.80 0.57 0.69
CLIP 0.80 0.82 0.67 0.68
SigLIP2 0.79 0.78 0.68 0.74
DINOv2 0.72 0.80 0.70 0.81
MAE 0.42 0.52 0.67 0.61
(c) Gain (M1 − M0)
RemoteCLIP 0.11 0.08 0.10 0.07
CLIP 0.10 0.21 0.09 0.13
SigLIP2 0.13 0.10 0.14 0.16
DINOv2 0.04 0.07 <0.01 0.09
MAE 0.21 0.31 0.21 0.11
Blur
Brightness
Noise
Block
Own-class argmax rate
1.00
0.75
0.50
0.25
0.00
M1 − M0
0.30
0.20
0.10
0.00